Chapter 10

Data Clustering

10.1 Introduction

The goal of data clustering/cluster analysis is to discover the “natural" grouping on a set of vector points plotted on an N dimensional space, where N represents the number of attributes by which the objects to be clustered is represented. I.e. the original dataset is partitioned into different subsets where data in each subset ideally shares some common traits. Data clusters are the product of an unsupervised classification of different patterns involving data item, observations or feature vectors.

Now cluster analysis is prevalent in any discipline that involves analysis or processing of multivariate data. Data clustering is used in many exploratory process including pattern-analysis, decision making, grouping. Cluster analysis is widely used in market research when working with multivariate data from surveys and test panels [3]. Market researchers use cluster analysis to partition the general population of consumers into market segments and to better understand the relationship between different groups of potential customer’s e.g. Product positioning, selection of test markets etc. Clustering of Web–log sessions helps to discover groups of access patterns which are similar with respect to a suitable similarity measure. These access patterns characterize different user groups. This kind of information can be used to develop marketing strategies.

It is also heavily used in machine learning applications like image segmentation, document retrieval, Climatology, Petroleum Geology, Crime Analysis, Search results grouping, data pattern identification etc [4]. Different writing styles were automatically discovered by clustering in to facilitate online handwriting recognition. Contours of MR brain images are clustered into different subclasses for easy diagnosis when a new data (MR brain image) arrives. Documents can be clustered to generate topical hierarchies for information access or retrieval. The study of genome data in biology often involves clustering -either on the subjects, the genes, or both.

The figure 10.1 (a) shows the figures obtained after plotting the different data set (two dimensional). The clusters in this data set, though easily identified by a human, are difficult to be detected automatically. The clusters are of different shapes, sizes and densities. Background noise makes the clustering task even more difficult. The figure 10.1 (a) shows random, spiral, elongated and circular datasets. Figure 10.1(b) shows the clustered output.[Refer the attached CD for original color image]

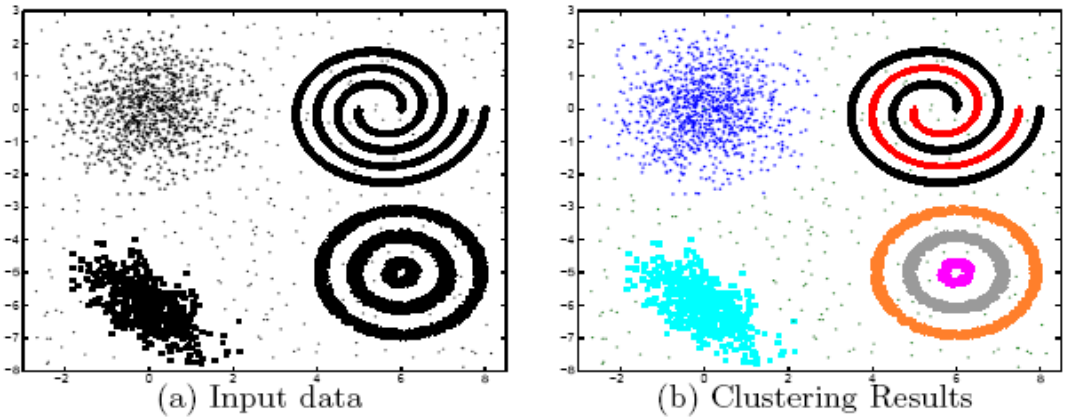


Fig 10.1: The datasets & identified clusters for data sets (Printed with permission from [2])

10.2 Types of Clustering

Many clustering algorithms have been proposed in different application scenarios. The two main categories of clustering algorithm are – the hierarchical clustering and partitiononal clustering.

10.2.1 Partitioning Methods

The partitioning methods generally result in a set of M clusters, each object belonging to one of the M clusters. Each cluster may be represented by a centroid or a cluster representative. [This is some sort of summary description of all the objects contained in a cluster.] The precise form of this description will depend on the type of the object which is being clustered. In case where real-valued data is available, the arithmetic mean of the attribute vectors for all objects within a cluster provides an appropriate representative; alternative types of centroid may be required in

other cases, e.g., a cluster of documents can be represented by a list of those keywords that occur in some minimum number of documents within a cluster. If the number of the clusters is large, the centroids can be further clustered to produces hierarchy within a dataset. This method works well for finding spherical- shaped clusters in small or medium sized databases.

Single Pass Partitioning: Thus is a very simple partition method; the single pass method creates a partitioned dataset as follows:

1. Make the first object the centroid for the first cluster.
2. For the next object, calculate the similarity, S, with each existing cluster centroid, using some similarity coefficient.
3. If the highest calculated S is greater than some specified threshold value, add the object to the corresponding cluster and re determine the centroid; otherwise, use the object to initiate a new cluster. If any objects remain to be clustered, return to step 2.

As its name implies, this method requires only one pass through the dataset; the time requirements are typically of order *O(NlogN)* for order *O(logN)* clusters. This makes it a very efficient clustering method for a serial processor. A disadvantage is that the resulting clusters are not independent of the order in which the documents are processed, with the first clusters formed usually being larger than those created later in the clustering run.

Important partitional clustering algorithms in the pattern recognition community includes, the k-means algorithm, PAM(Partitioning around Medoids), CLARA(Clustering Large Applications), K-mode, the EM algorithm, CLARANS(Clustering Large Applications based on Randomized search), different types of linkage methods, the mean-shift algorithm, algorithms that minimize some graph-cut criteria, path-based clustering and different flavors of spectral clustering.

10.2.2 Hierarchical methods

In hierarchical clustering the data are not partitioned into a particular cluster in a single step. Instead the algorithms work by finding successive clusters with the use of previously

established clusters. Hierarchical algorithms can be further subcategorized as an agglomerative methods ("bottom-up"), which proceeds by series of fusions of the n objects into groups or divisive methods ("top-down"), which proceeds by separating n objects successively into finer groupings. Hierarchical clustering is usually represented by a two dimensional diagram known as dendrogram as shown in figure 10.2. It illustrates the fusion or division made at each level or stage of the cluster analysis. The diagram in figure 10.2 show the general structure of a dendrogram [5].

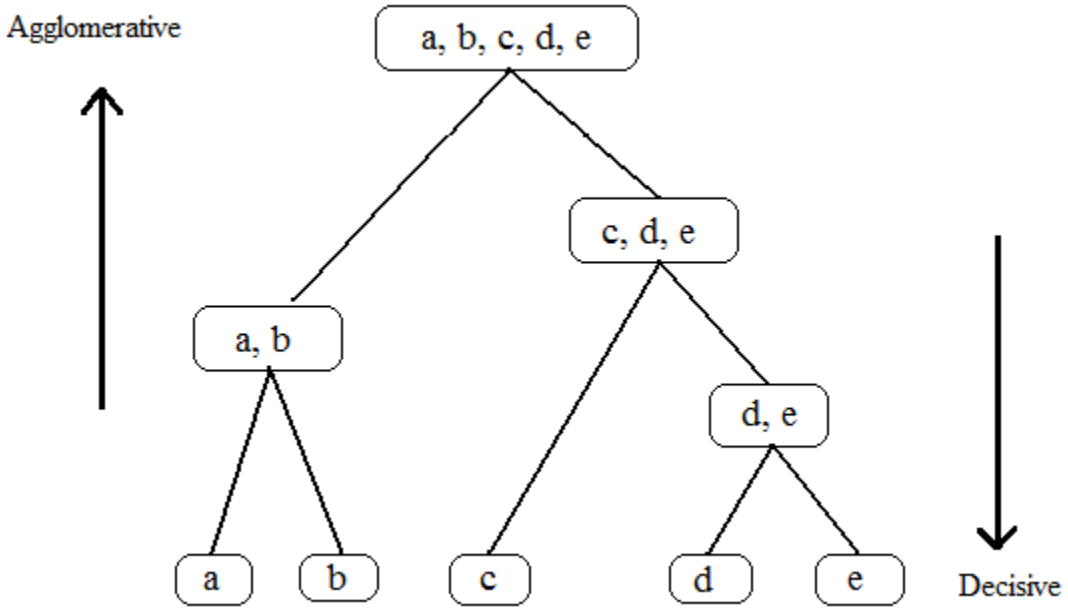


Fig 10.2: An example dendrogram.

10.2.2.1 Hierarchical Agglomerative Clustering

The agglomerative approach starts with each pattern in a distinct cluster (singleton) and then successively does merging of rest of the data until a certain condition is being met. The construction of a hierarchical agglomerative classification can be achieved by the following general algorithm.

1. Find the two closest objects and merge them into a single cluster

2. Find and merge the next two closest points, where a point is either an individual object or a cluster of objects.
3. If more than one cluster remains , return to step 2

Individual methods are characterized by the definition used for identification of the closest pair of points, and by the means used to describe the new cluster when two clusters are merged.

10.2.2.2 Hierarchical Divisive Classification

The divisive approach starts with all clusters patterns within a single cluster and then splits them until a condition is satisfied. The construction of hierarchical divisive classification can be achieved by the following algorithm

1. Assume the whole dataset as the single cluster.
2. Now split the cluster using a flat clustering algorithm until the required number of clusters is reached [5][6].

10.2.3 Density Based Methods

This method groups nearby objects into clusters based on density conditions. It either grows cluster according to the density of neighborhood or according to some density functions. The DBSCAN (Density based spatial clustering of Applications with Noise), GDBSCAN (Generalized Density Based Spatial Clustering of Applications with Noise), OPTICS (Ordering Points To Identify the Clustering Structure), LOF (Local Outlier Factors) are the major algorithms which fall under this category [7].

NOTE

DBSCAN is applied to a 5D feature space created from 5 different spectral channels obtained from several satellite images covering the area of California to detect the *landuse* data for the region.

10.2.4 Grid Based Methods

The methods quantize the space into a finite number of the cells that form a grid structure and then perform clustering on the grid structure. The STRING (Statistical Information Grid-based method) and the WAVE CLUSTER methods are the two major additions from this class of clustering algorithms [1].

10.2.5 Model Based Methods

The methods hypothesize a model for each of the clusters and find the best fit of the data to that model. Typical model-based methods involve statistical approaches (COBWEB, CLASSIT, AutoClass) or neural network approaches (Competitive Learning and Self Organizing maps) [1].

Universal Clustering Algorithm - An elusive goal!

Now you might be thing of why, should we have these many algorithms for clustering. This is because the existing clustering techniques do not address all the clustering requirements adequately and concurrently. Each algorithm uses different methods to achieve clustering and the clusters which we receive at the end will be different. Thus a universal clustering algorithm has always remained an elusive goal. The fundamental reason for this is the intrinsic ambiguity of the notion of natural grouping [1].

Another difficulty is the diversity of clusters: clusters can be of different shapes, different densities, different sizes, and are often overlapping. The problem is even more challenging when the data is high-dimensional, because the presence of irrelevant features can obscure the cluster structure. Above all, even for data without any cluster structure, most clustering algorithms still generate spurious clusters. All these issues make clustering a dilemma for the user .

Note

The hierarchical agglomerative clustering methods are most commonly used clustering technique.

10.3 Subjectiveness of Clustering

The clustered output/clusters formed are highly dependent on the selected parameters/features of the objects to be clustered. For example, take the case in figure 10.3. Here we have the nine people whom we have to group. But the first question which comes to our mind is: On what basis they have to be grouped?

Here the grouping can done based on basis of family relations (see Fig 3.5 (a)), sex of individual (see Fig 3.5 (c)), color of skin etc. And for each of these basis we obtain a different cluster combination. Thus it is true to say that “Clustering is Subjective”.

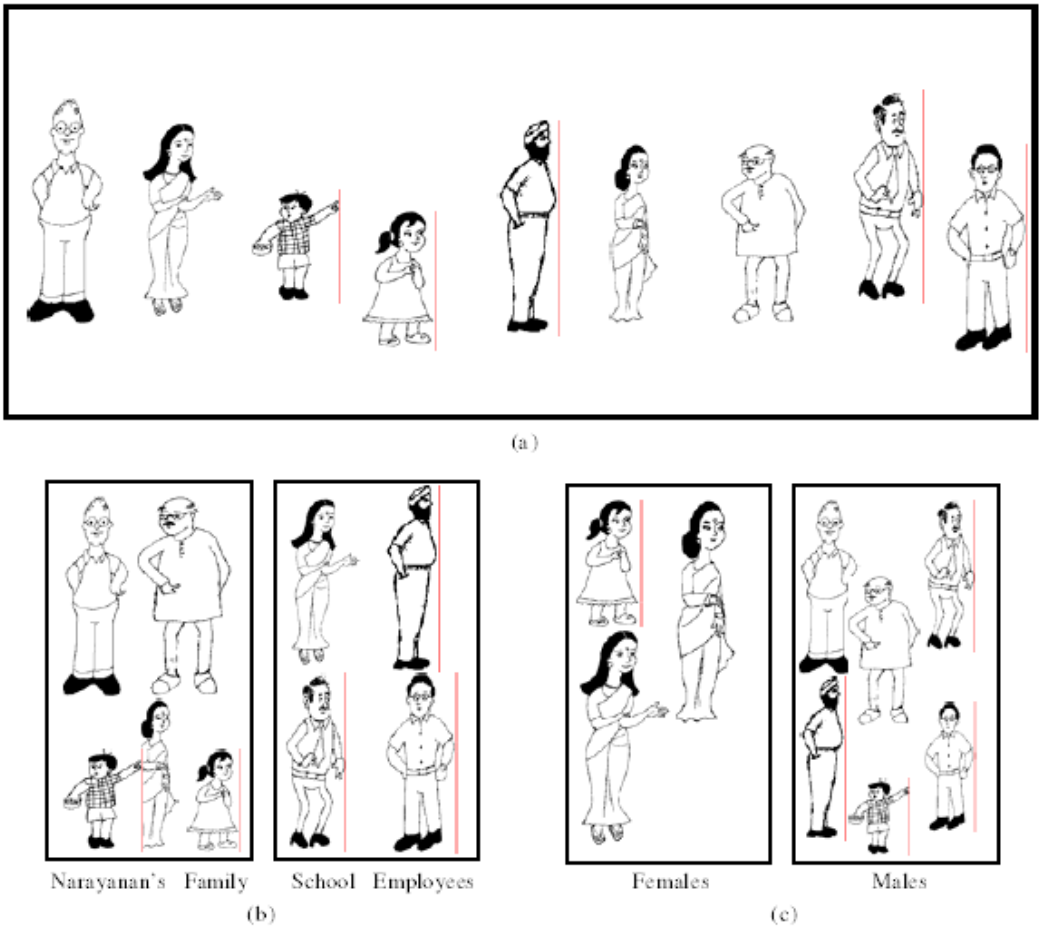


Fig 10.3: Clustering is subjective (printed with permission from [1])

In most practical situations of cluster analysis, the investigator knows enough about the problem to distinguish good grouping from bad groupings [1].

10.4 Cluster Dissimilarity/Similarity

In order to decide which clusters should be combined (for agglomerative), or where a cluster should be split (for divisive), a measure of dissimilarity between sets of observations is required. In most methods of hierarchical clustering, this is achieved by use of an appropriate metric (a measure of distance between pairs of observations), and a linkage criterion which specifies the dissimilarity of sets as a function of the pairwise distances of observations in the sets.

Metric

The choice of an appropriate metric will influence the shape of the clusters, as some elements may be close to one another according to one distance and farther away according to another. For example, in a 2-dimensional space, the distance between the point (1,0) and the origin (0,0) is always 1 according to the usual norms, but the distance between the point (1,1) and the origin (0,0) can be 2, $\sqrt{2}$or 1 under Manhattan distance, Euclidean distance or maximum distance respectively.

Some commonly used metrics for hierarchical clustering are [3]:

Names	Formula
Euclidean distance	$\|a-b\|_2 = \sqrt{\sum_i (a_i - b_i)^2}$
squared Euclidean distance	$\|a-b\|_2^2 = \sum_i (a_i - b_i)^2$
Manhattan distance	$\|a-b\|_1 = \sum_i \|a_i - b_i\|$
maximum distance	$\|a-b\|_\infty = \max_i \|a_i - b_i\|$
Mahalanobis distance	$\sqrt{(a-b)^\top S^{-1}(a-b)}$ where S is the covariance matrix
cosine similarity	$\frac{a \cdot b}{\|a\|\|b\|}$

Fig 10.4: Distance Metrices

Among this the most commonly used distance measure in published studies in that research area is the Euclidean distance or the squared Euclidean distance. For text or other non-numeric data, metrics such as the Hamming distance or Levenshtein distance are often used.

10.5 K-Means Clustering Algorithm

K-Means is the simplest partitional clustering algorithm. For explaining this, let us define our data points $\boldsymbol{x}_i \in \mathbb{R}^n$ where in a given sample i will range from 1 to m. We make our first assumption that there are at most K possible groupings or clusters within the data. Of course what leads us to make this assertion will be application specific but clearly by now you should be able to see that this value K is one over which we would wish to do inference. However for the simplest of algorithms it is considered that K - the number of clusters - is given a priori. We will also assume that there are a number of binary indicator variables associated with each data point and cluster z_{ki} which takes on the value 1 if $\boldsymbol{x}_i$ has been assigned to cluster k and zero otherwise [2].

10.5.1 Measure of Cluster Quality

Now we need to define a measure which will indicate the quality of the clustering. Once this is defined it can be employed in guiding the selection of the optimal clustering of our data. We would be more convinced that a cluster which had been identified was not some spurious random artifact if the internal cohesiveness of the points allocated to the cluster was high. A simple way of defining this is to measure how close each of the cluster points is to the cluster average. If all the points are tightly surrounding the mean then the cluster is highly cohesive. On the other hand if the data points are diffusely spread around the cluster centre then our confidence in the cluster actually representing a compact grouping of self-similar points will be low. Given this we can define a measure of cluster compactness as the total distance from the cluster mean in other words

$$\sum_{\boldsymbol{x}_i \in C_k} \left\| \boldsymbol{x}_i - \overline{\boldsymbol{x}}_k \right\|^2 = \sum_{i=1}^{m} z_{ki} \left\| \boldsymbol{x}_i - \overline{\boldsymbol{x}}_k \right\|^2 \tag{10.1}$$

where the cluster mean is defined as $\bar{\boldsymbol{x}}_k = \frac{1}{m_k} \sum_{\boldsymbol{x}_i \in C_k} \boldsymbol{x}_i$ and $m_k = \sum_{i=1}^{m} z_{ki}$ is the total number of points allocated to cluster k. The total goodness of the clustering will then be based on the sum of the cluster compactness measures for each of the K clusters. Using the indicator variables z_{ki} then we can define the overall cluster goodness as

$$\varepsilon_K = \sum_{i=1}^{m} \sum_{k=1}^{K} z_{ki} \left\| \boldsymbol{x}_i - \bar{\boldsymbol{x}}_k \right\|^2 \tag{10.2}$$

So we have our overall measure of cluster quality the next step is to devise an algorithm which will allow us to optimize this [2].

10.5.2 Iterative Criterion Optimization Algorithm

The criterion which we have to optimize is $\varepsilon_K = \sum_{i=1}^{m} \sum_{k=1}^{K} z_{ki} \left\| \boldsymbol{x}_i - \bar{\boldsymbol{x}}_k \right\|^2$. There are two sets of parameters - the cluster mean values $\bar{\boldsymbol{x}}_k$ and the cluster allocation indicator variables z_{ki}. We will optimize our criterion over each set of variables by holding one set fixed whilst performing optimization over the other set. So given our current indicator variables z_{ki} the optimal value of the mean vectors $\bar{\boldsymbol{x}}_k$ simply boils down to the estimates based on the data points allocated to each cluster. Convince yourself that this is the case by differentiating the cluster criterion with respect to each $\bar{\boldsymbol{x}}_k$ and solving. Therefore given each z_{ki} we obtain our K-means by $\bar{\boldsymbol{x}}_k = \frac{\sum_{i=1}^{m} z_{ki} \boldsymbol{x}_i}{\sum_{i=1}^{m} z_{ki}}$ Now give each of our new $\bar{\boldsymbol{x}}_k$ we need to update the values of our indicator values z_{ki}. From the expression for ε_K we can see that each $\boldsymbol{x}_i$ should be assigned to the cluster k for which it has the shortest distance to the cluster centre i.e. $\left\| \boldsymbol{x}_i - \bar{\boldsymbol{x}}_k \right\|^2$ is the smallest. This is then just a simple matter of finding the centre with the minimum distance to the data point under question and setting the associated indicator variable to this cluster value i.e z_{ki} = 1 for k which yields the minimum of $\left\| \boldsymbol{x}_i - \bar{\boldsymbol{x}}_k \right\|^2$.

Once these values have been redefined then we can go back and revise our estimates of each $\bar{x}_k$ and continue this iteration until ε_K converges to some steady value. This is very simple algorithm and is the K-Means Clustering algorithm for which a simple Matlab implementation is included here [2].

```
function [M,j,e] = kmeans(X,K,Max_Its)
[m,n]=size(X);
I=randperm(m);
M=X(I(1:K),:);
Mo = M;
        for n=1:Max_Its
                        for k=1:K
                        Dist(:,k) = sum((X - repmat(M(k,:),m,1)).^2,2)';
                        end
                [i,j]=min(Dist,[],2);
                        for k=1:K
                        if size(find(j==k))>0
                        M(k,:) = mean(X(find(j==k),:));
                        end
        end
        Z = zeros(m,K);
                for p=1:m
                        Z(p,j(p)) = 1;
                end
                e = sum(sum(Z.*Dist)./m);
                fprintf('%d Error = %f\n', n, e);
                Mo = M;
        end
end
```

10.5.3 Implementation of k-means clustering In Microsoft Excel

The above mentioned Matlab code shows the implementation of K-Means clustering algorithm. Implementing the same in Microsoft Excel would make the algorithmic concept more understandable. The implementation of K-Means in Excel is possible with the help of Solver and What-If Analysis available with Microsoft Excel 2007 and later versions.

Prerequisite

For demonstrating the working of k means algorithm, it would be better if we have a set of data points which are suitably selected so that clusters can easily be distinguished. The table shown in figure 10.5.1 shows one such data set of data points from which three clusters can be easily distinguished.

	B	C	D
5	TEST DATA		
6	SL NO	X	Y
7	1	4	12
8	2	5	10
9	3	8	7
10	4	5	3
11	5	5	4
12	6	2	11
13	7	5	4
14	8	3	8
15	9	6	2
16	10	7	4
17	11	10	8
18	12	8	9
19	13	10	9
20	14	10	12

Fig 10.5.1: Test Data

Form the figure 10.5.2 it is possible to easily distinguish three clusters (k=3). Now our objective is to use K-means algorithm and find out the three cluster group present in the data distribution.

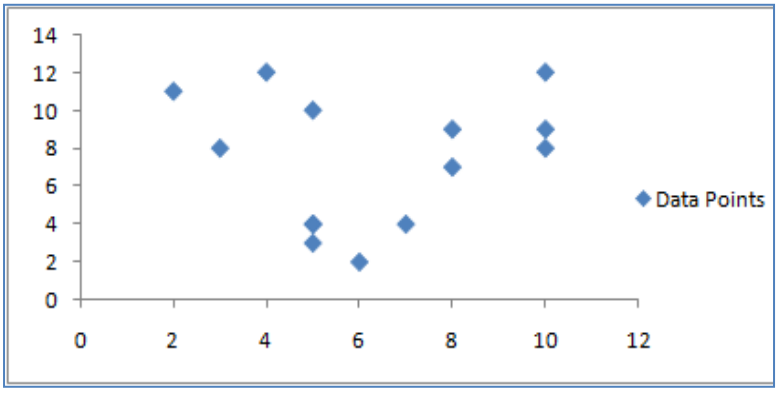


Fig 10.5.2: Plot of test data points

The K–Means algorithm starts with the assumption of K .Here we have assumed three clusters (K = 3). Now the three cluster centers are initialized with random points. Lets this be (4, 4), (5, 12), (10, 6). This is entered in the separate cells as shown in the figure 10.5.3(a).

	F	G	H	I	J	K
6	Cluster Center 1		Cluster Center 2		Cluster Center 3	
7	X	Y	X	Y	X	Y
8	4	4	5	12	10	6

Fig 10.5.3 (a): Initial assumption of cluster centers

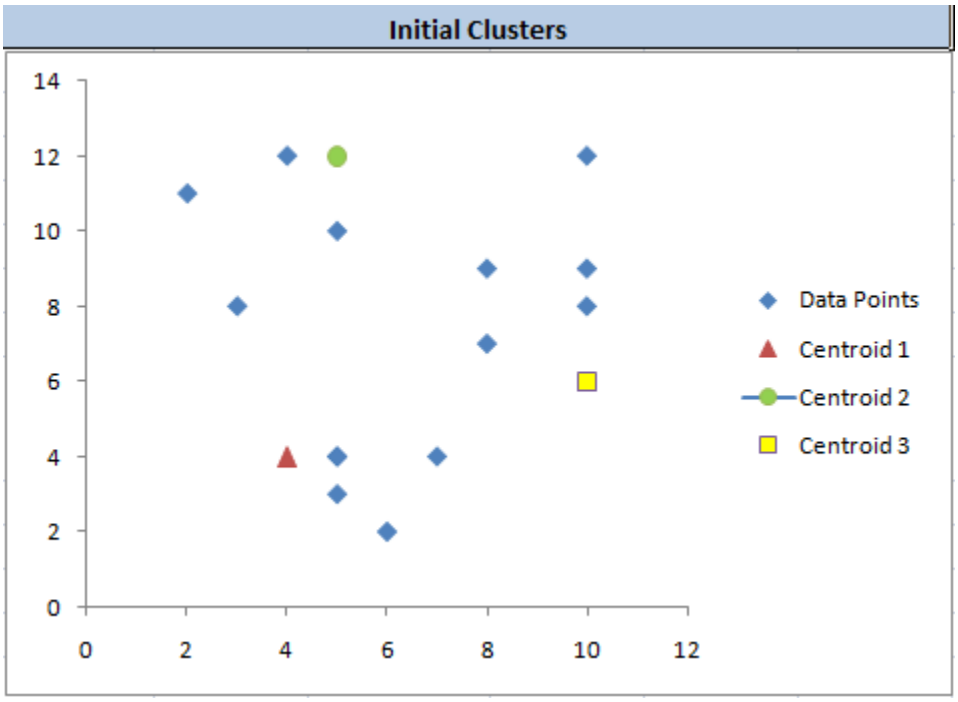


Fig 10.5.3 (b): Initial assumption of cluster centers(plotted)

Our objective here is to minimize $\sum_{j=1}^{k}\sum_{i=1}^{n}\left\|X_i^{(j)} - C_j\right\|^2$, where X_i is the data point belonging to the cluster and C_j is the cluster center (centriod). So as the next step, we need to find out the eucledian distance of the each data point with all the centroids. For this here, we utilize the concept of 'What-If Analysis' feature available with excel. As a prerequisite we need to make available a structure as shown

in figure 10.1.4, where the X and Y values for the chosen index will be taken from the table shown in figure 10.5.1.

	G	H	I	J
10	INDEX OF CURRENT POINT		X	Y
11	1		4	12

Fig 10.5.4: A requirement for what if analysis

For using 'What-If Analysis' the columns values 1 to 14 are written in one column as shown in screenshot (figure 10.5.5 of the table). In the first row of the second column, type the formula to find the eucledian distance of centroid 1 from data point 1. Eg: =SQRT(($J$11-$G$8)^2+($K$11-$H$8)^2). $G$8, $G$8 refers to the, X coordinate of the centroid 1 (X column value of the table in figure 11.5.4). $K$11, $H$8 refers to the, Y coordinate of the centroid 1 (Y column value of the table in figure 13.1.4). In the third column type the formula for finding distance of second centroid with the data point 1. In the next column do find the distance with the third centroid. In the last column find out the class to which the data point 1 belongs. The formula used to obtain this is =IF(MIN(O7:Q7)=O7,"Class1",IF(MIN(O7:Q7)=P7,"Class2","Class3")). Here MIN(O7:Q7) takes the minimum distance from the distances calculated for the data point 1 ,2 and 3. The formula also implements the functionality finding out from which column the minimum value comes. ie. If the minimum comes from the second column we assign the point to "Cluster 1", if it comes from the third column we assign this point as a "Cluster 2" point and else if it comes from the fourth column, we assign "Cluster3". This is done in order to assign the point to a particular cluster. The 'What-If Analysis' will do the above mentioned operations for all the point by changing the index value of the table shown in figure 10.5.5 from 1 to 14.

Now we need to find out the minimum distance of each point from their each cluster centers. This calculation is what is being done in the next column. In the column having label 'Minimum distance" we calculate the sum of minimum distances of all the data point. Now we calculate the sum of the above calculated minimum distance values for all the points as shown

in figure 10.5.5. [The minimum distance calculation is done twice just for the sake of easiness of understanding].

	M	N	O	P	Q	R	S	T
6	INDEX	Distance to CENTROID 1	Distance to CENTROID 2	Distance to CENTROID 3	X	Y	CLASS	Minimum Distance
7	1	8	1	8.485281374	4	12	Cluster2	1
8	2	6.08276253	2	6.403124237	5	10	Cluster2	2
9	3	5	5.830951895	2.236067977	8	7	Cluster3	2.236067977
10	4	1.414213562	9	5.830951895	5	3	Cluster1	1.414213562
11	5	1	8	5.385164807	5	4	Cluster1	1
12	6	7.280109889	3.16227766	9.433981132	2	11	Cluster2	3.16227766
13	7	1	8	5.385164807	5	4	Cluster1	1
14	8	4.123105626	4.472135955	7.280109889	3	8	Cluster1	4.123105626
15	9	2.828427125	10.04987562	5.656854249	6	2	Cluster1	2.828427125
16	10	3	8.246211251	3.605551275	7	4	Cluster1	3
17	11	7.211102551	6.403124237	2	10	8	Cluster3	2
18	12	6.403124237	4.242640687	3.605551275	8	9	Cluster3	3.605551275
19	13	7.810249676	5.830951895	3	10	9	Cluster3	3
20	14	10	5	6	10	12	Cluster2	5
21							Sum of Minimum distances	35.36964323

Fig 10.5.5: What if analysis/Distance calculation for individual clusters

Now we need to use “SOLVER” to minimize sum of, sum of distances to the nearest centroid points. The below figure 10.5.6 depicts on, how to the find the optimal cluster centers using solver. The range of the objective function and the changeable cells are being provided in the figure 10.5.6.

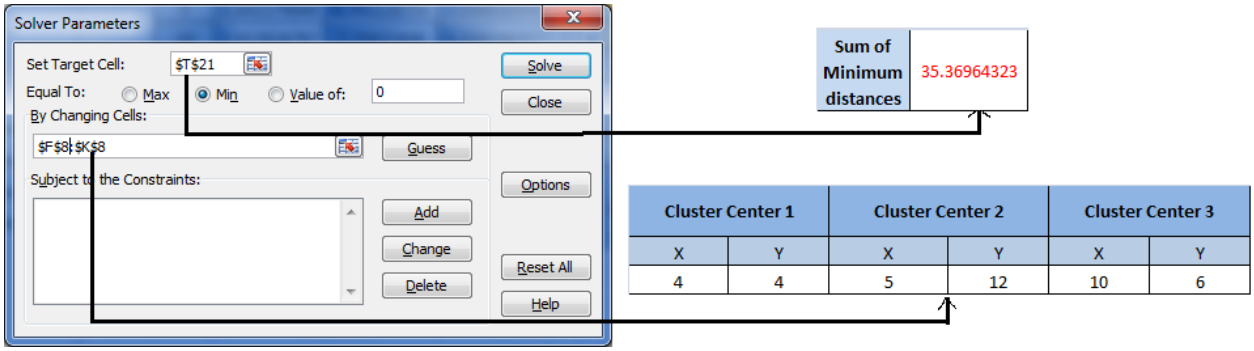


Fig 10.5.6: Providing Solver parameters for Cluster1

On clicking the “Solve’ button, the optimal values for all three centroids are obtained. The Figure 10.5.7 shows the minimized distances value and optimal cluster centers. These cluster centers will come very close to the center of each cluster. This is shown in figure 10.5.8.

Sum of Minimum distances	21.01111085

(a) Minimized Euclidian distance

	F	G	H	I	J	K
6	Cluster Center 1		Cluster Center 2		Cluster Center 3	
7	X	Y	X	Y	X	Y
8	5.21815702	3.66736723	3.615383398	10.4615335	9.47841363	8.75055097

(b) Optimal cluster centers

Fig 10.5.7

We can then plot the final point which are the optimal cluster centers as show in the figure 10.5.7. Here we can see that the cluster centers have moved to optimal locations i.e. close to the center of the initially assumed clusters. The final position of cluster centers is shown in figure 10.5.8.

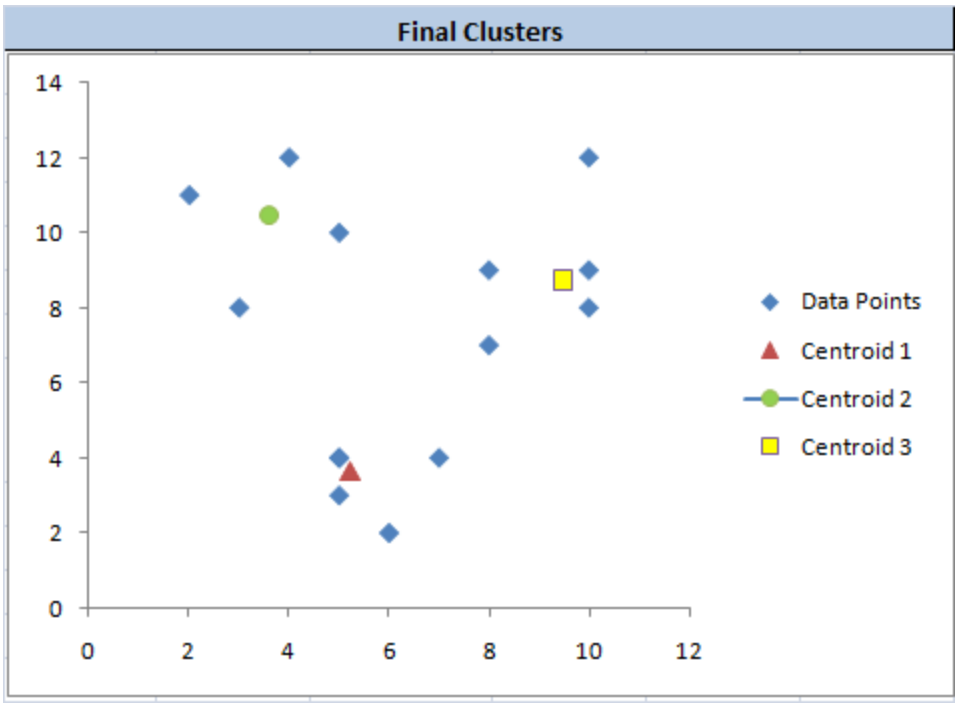


Fig 10.5.8: Final position of cluster centers

10.6 K- Medoids Clustering Algorithm

Both K-means and k-medoids have similar procedures. In k-medoids algorithm, only data points in the space can become Medoids; however in the k-means algorithm any point in the space near the data points or data points themselves can be mean point. Based on the cost calculated between a point and an assumed medoid, the points are swapped or retained as Medoids until there are no net changes for all points for the medoids assumed [1].

K-medoid is a typical partitioning algorithm. The objective of using this algorithm is , for a given k; find k representatives in the dataset so that, when assigning each object to the closest representative, the sum of distances between representatives and object, which are assigned to them, is minimal .

10.6.1 Implementation of K-Medoids in Microsoft Excel

The implementation of k-medoids in Microsoft Excel is not as easy as k-means as k-medoids has a unique requirement that, the cluster centers should be one of the data points. But this

requirement cannot be directly taken care of by ‘solver’ and hence we avoid the use of ‘solver’ and go in a roundabout way in this case. This is explained in detail below.

Prerequisite

As one can see in the algorithm, K- medoids is not much different from the original algorithm. Here also as a prerequisite we require test data points which can be clustered easily. The below figure 10.6.1 shows the plot of assumed data points. Here in order to satisfy the above mentioned requirement of k-medoids the indices of data points are utilized to do the iteration. Figure 10.6.3 shows the initially assumed indices. In each of the iteration the indices are updated with the newly found ones.

	C	D	E
5	TEST DATA		
6	SL NO	X	Y
7	1	3	13
8	2	3	10
9	3	8	7
10	4	5	2
11	5	5	4
12	6	2	11
13	7	5	4
14	8	2	14
15	9	6	2
16	10	4	2
17	11	10	8
18	12	8	9
19	13	10	9
20	14	10	12

Fig 10.6.1: Test data points

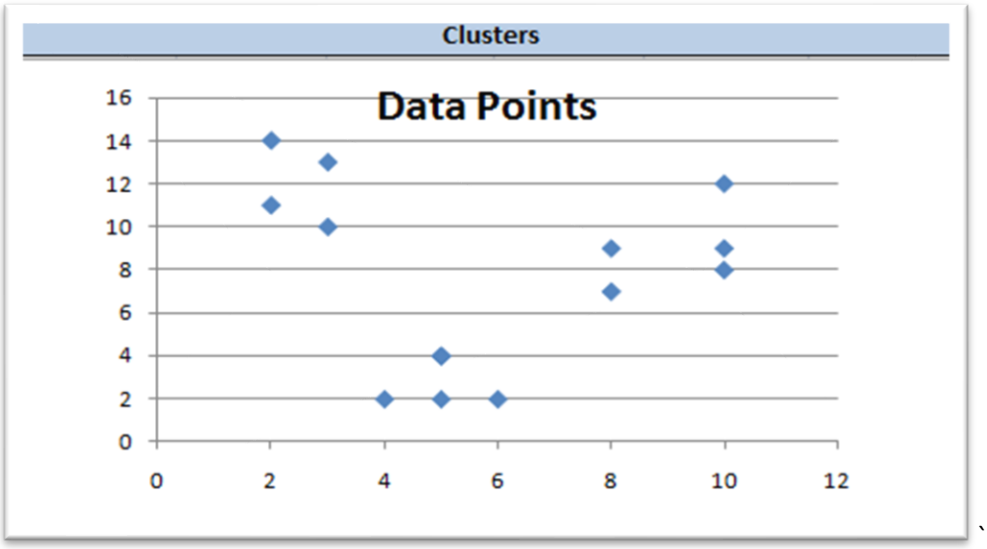


Fig 10.6.2: Plot of assumed data points

According to the K-medoids algorithm, the centroid should be a data point. For this reason we start with assuming three data points as the centroids. So we assume three indices 12, 1, 10 as shown in the figure (figure 10.6.3). The figure 10.6.4 shows the data points corresponding to the selected indices. The formula used in the cells labeled X and Y in the Figure 10.6.4 is =INDEX($D$7:$D$20,H7,1), =INDEX($E$7:$E$20,H7,1) where $D$7:$D$20 represents the range of cells shown in Figure 10.6.1 and H7 represents the location of first index value assumed(Here it is first cell value in figure 10.6.1 [The INDEX function will select the values from the specific row of the selected range].

	H	I	J
6	INDEX 1	INDEX 2	INDEX 3
7	12	1	10

Fig 10.6.3: Assumed Indices

	G	H	I	J	K	L
9	Cluster Center 1		Cluster Center 2		Cluster Center 3	
10	X	Y	X	Y	X	Y
11	8	9	3	13	4	2

Fig 10.6.4: Data points corresponding to assumed indices

	H	I	J	K
13	INDEX 1		X	Y
14	1		3	13

Fig 10.6.5: A requirement for “what if analysis”

For “what if analysis” the columns values 1 to 14 are written in one column as shown in below screenshot of the table. In the first row of the second column type the formula to find the eucledian distance of centroid 1 from data point 1. Eg: = SQRT(($J$14-$G$11)^2+($K$14-$H$11)^2). $J$14, $G$11 refers to the ,X coordinate of the centroid 1 (corresponding to X label of the figure 10.6.5). $K$14, $H$11 refers to the, Y coordinate of the centroid 1 (corresponding to Y label of the figure 10.6.5). In the third column type the formula for finding distance of second centroid with the data point 1. Do this with the third centroid also. For computation purpose, store the X and Y values displayed for a particular index (as shown in figure 10.6.5) in its next 2 column. In the last column find out the class to which the data point 1 belongs. The formula used to obtain this is - =IF(SMALL(O7:Q7,1)=O7,"Cluster1",IF(SMALL(O7:Q7,1)=P7,"Cluster2","Cluster3")) here SMALL(O7:Q7,1) takes the minimum distance from the distances calculated for the data point 1 ,2 and 3. The formula also implements the functionality finding out from which column the minimum value comes. ie. If the minmum comes from the second colum we assign the point to “Cluster 1”, if it comes from the third column we assign this point as a “Cluster 2” point and else if it comes from the fourth column, we assign “Cluster3”. This is done in order to assign the point to a particular cluster. The “what if analysis “ will do the above mentioned operations for all the point by changing the index value of the table shown in figure 10.6.6 from 1 to 14.

	N	O	P	Q	R	S	T
6	INDEX	Distance to CENTROID 1	Distance to CENTROID 2	Distance to CENTROID 3	X	Y	CLASS
7	1	6.403124237	0	11.04536102	3	13	Cluster2
8	2	5.099019514	3	8.062257748	3	10	Cluster2
9	3	2	7.810249676	6.403124237	8	7	Cluster1
10	4	7.615773106	11.18033989	1	5	2	Cluster3
11	5	5.830951895	9.219544457	2.236067977	5	4	Cluster3
12	6	6.32455532	2.236067977	9.219544457	2	11	Cluster2
13	7	5.830951895	9.219544457	2.236067977	5	4	Cluster3
14	8	7.810249676	1.414213562	12.16552506	2	14	Cluster2
15	9	7.280109889	11.40175425	2	6	2	Cluster3
16	10	8.062257748	11.04536102	0	4	2	Cluster3
17	11	2.236067977	8.602325267	8.485281374	10	8	Cluster1
18	12	0	6.403124237	8.062257748	8	9	Cluster1
19	13	2	8.062257748	9.219544457	10	9	Cluster1
20	14	3.605551275	7.071067812	11.66190379	10	12	Cluster1

Fig 10.6.6: What if analysis Data Table

Now we need to find the mean values for X and Y coordinates for those set of points which are assigned to one Cluster. This is done for all the clusters. The screen shot below shows how this is implemented.

	V	W	X	Y	Z	AA
6	Average of Custrer 1 points(X)	Average of Custrer1 points(Y)	Average of Custrer 2 points(X)	Average of Custrer 2 points(Y)	Average of Custrer 3 points(X)	Average of Custrer 3 points(Y)
7	0	0	3	13	0	0
8	0	0	3	10	0	0
9	8	7	0	0	0	0
10	0	0	0	0	5	2
11	0	0	0	0	5	4
12	0	0	2	11	0	0
13	0	0	0	0	5	4
14	0	0	2	14	0	0
15	0	0	0	0	6	2
16	0	0	0	0	4	2
17	10	8	0	0	0	0
18	8	9	0	0	0	0
19	10	9	0	0	0	0
20	10	12	0	0	0	0
21	**9.2**	**9**	**2.5**	**12**	**5**	**2.8**

Fig 10.6.7; Implements logic for finding mean values of Clusters

	V	W	X	Y
23	INDEX	Distance of Centroid 1 with other points	Distance of Centroid 2 with other points	Distance of Centroid 3 with other points
24	1	7.378346698	1.118033989	10.39422917
25	2	6.280127387	1.346291202	7.472616677
26	3	1.943650632	7.340905182	5.16139516
27	4	8.163332653	10.0277393	0.666666667
28	5	6.529931087	8.117197108	1
29	6	7.472616677	0.559016994	8.731551981
30	7	6.529931087	8.117197108	1
31	8	8.7658428	1.767766953	11.59482643
32	9	7.696752562	10.33870828	1.067187373
33	10	8.720091743	9.809292646	1.067187373
34	11	1.067187373	8.498365856	7.213875519
35	12	1	6.262764743	6.887670143
36	13	0.666666667	8.117197108	7.964923101
37	14	2.587362449	7.673909622	10.47091209
38	Index of MINIMUM DISTANCE Data Point	**13**	**6**	**4**

Fig 10.6.8: Implements logic for finding index of data point close to the avea

As the next step we copy the new indices and paste it at the location where the initial assumptions of indices were made. Once the initial assumption of centoid changed, the output also changes. Now the ouput indices are copied back as input. This process it repeated until the output continues to remain constant for previous few iteration. We then plot the final point which are the optimal cluster centers as show in the figure 10.6.9. The iteration process mentioned is achieved by using a macro code. The code is shown below.

```
Sub getCentroid()
N=20
For i = 1 To N
Sheet1.Range("W38:Y38").Copy
Sheet1.Range("H7:J7").PasteSpecial (xlPasteValues)
Next i
End Sub
```

The code implements the logic of copying the new indices obatined values to the location where the initial assumption of indices are made. At the end of 20 iterations or less the ouput will converge to the centroid point. Here the centroid point will be one of the data points.[see attached CD for color image of the below figure].

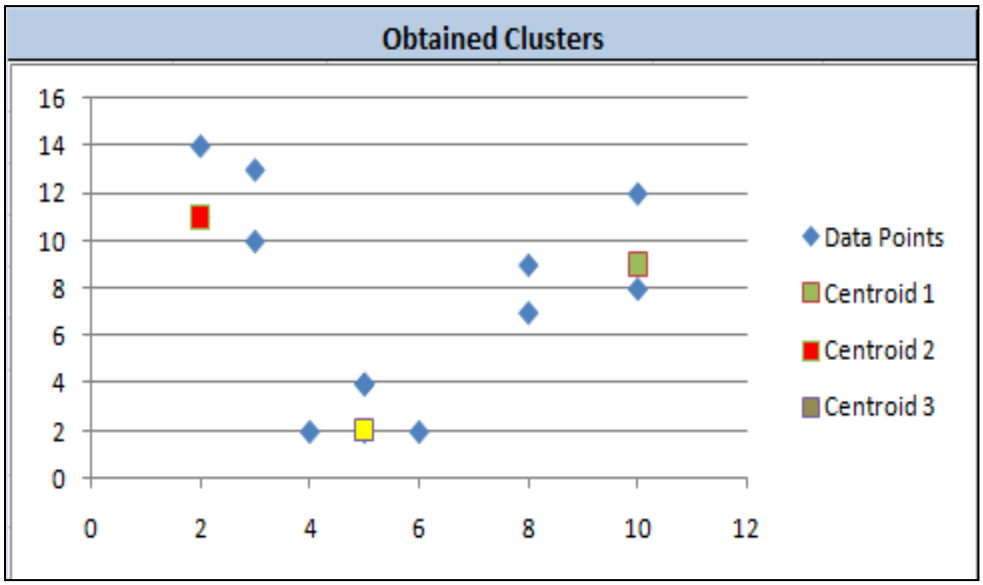


Fig 10.6.9: Centrioid point obtained for different clustes

10.7 Applications of clustering

10.7.1 Color Image Segmentation

A color picture is made of three components R (Red), G (green) and B (components). As mentioned in the previous example clustering basically groups the different similar data points, depending on the euclidian distance of its property values in property space. Here we have selected a color image and has clusterd different colors in a picture. The number of clusters created will be dependent on the value of k. Here we have chosen K = 4. Due to the easiness to implement here we have chosen K-means algorithm for color component clustering.

Obtaining pixel values from image

For segmentation the primary requirement would be to read the image to be segmented into Microsoft Excel sheet. Reading a color image involves, reading the Red, Green and Blue pixel components of the

image. This was initially not possible with Microsoft excel. Hence we developed a Microsoft Excel add-in to do the same (developed in C#). This will give the corresponding pixel values (Red, Green and Blue pixel values) for the image. Reading of image is done well within excel by using a user defined function named 'loadImageArray' availed from the add-in. The add-in once installed can be obtain from the functions tab of Microsoft Excel. [The add-in is available along with the attached CD].

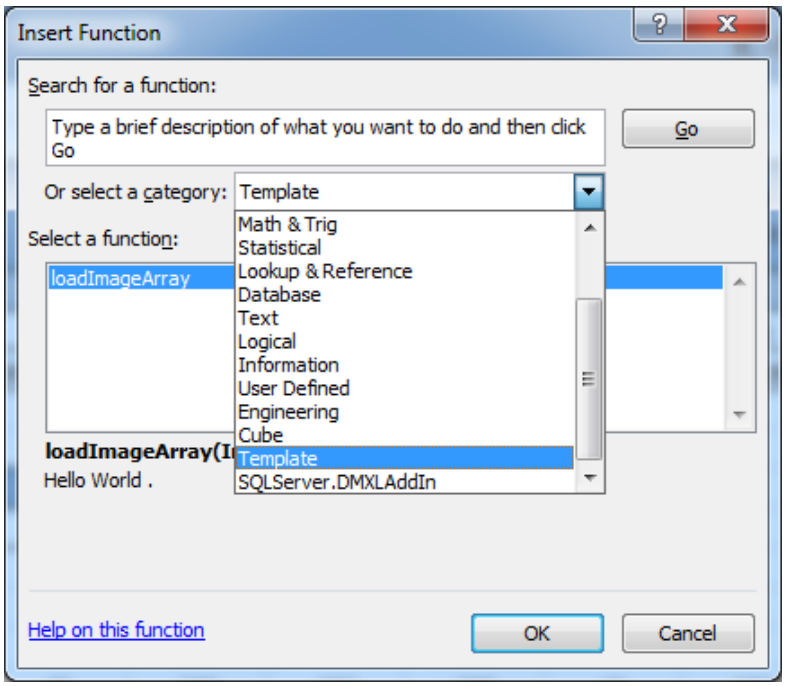


Fig 10.7.1: Image description of selecting "loadImageArray"

Once you select the template, we can see "loadImageArray" popping up in the function list. On selecting this function one can see the below given popup window. It will ask for four arguments – Location of image (should be given within double quotes), length of image, width of image and finally the color component you are interested in. ie. Give 1 for red, 2 for green and 3 for loading blue pixels. [The main objective of the add-in was to enable Microsoft Excel to obtain image pixel values without the help of third party software].

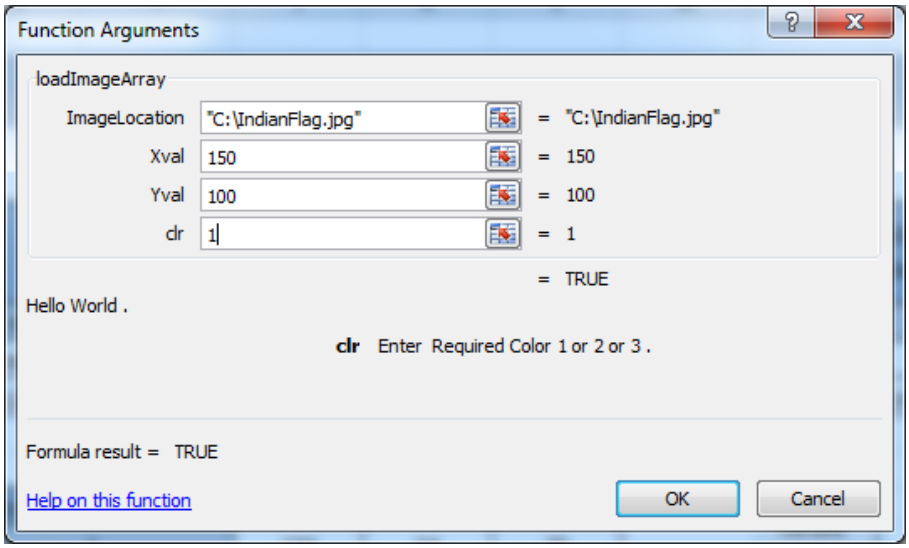


Fig 10.7.2: "loadImageArray" - details input screen

On clicking "ok", the first pixel will be shown in the selected cell of the excel sheet. Beginning from that cell select a range 150 x 100 (for this example) and then press and then to go to the beginning cell press "Ctrl + Alt + Enter". Now we will get all the pixel value for red component. Do this same process to obtain the green and blue component by giving 2 and 3 in place of 1. As a second step we can plot this picture(Indian Flag). This can be done by suitably coloring the corresponding cells of an excel sheet by the color we obtain by the pixel value combination(R, G and B) which we have already obtained. We have achieved this coloring by using a macro. (To see the macro – see implementation in attached CD).



Fig 10.7.3: Indian Flag

We now have the pixel values for R, G and B for the image. To segment the image we need to randomly select four cluster centers (which here are four sets of R, G and B values). The values selected should be within the range [0 , 255] since pixel values can only be in this range. Here since we have selected four cluster centers, k=4. Let this be (66, 76, 45), (240, 230, 44), (44, 113, 110) and (5, 5, 200). This is entered in the separate cells as shown in the figure 10.7.4 below.

	H	I	J	K	L	M	N	O	P	Q	R	S
6	Cluster Center 1			Cluster Center 2			Cluster Center 3			Cluster Center 4		
7	R	G	B	R	G	B	R	G	B	R	G	B
8	66	76	45	240	230	230	44	113	110	5	5	200

Fig 10.7.4: Randomly selected Cluster centers

	K	L	M	N	O
11	INDEX OF CURRENT POINT		R	G	B
12	1		156	54	76

Fig 10.7.5: A requirement for what if analysis

The testdata is obtained by writing the pixel values obtained for the image in a single column (all red, green, blue should be in different columns). This is shown in below figure 10.7.6.

	C	D	E	F
5	TEST DATA			
6	SL NO	R	G	B
7	1	156	54	76
8	2	137	28	49
9	3	149	23	44
10	4	158	23	40
11	5	157	21	35
12	6	163	22	38
13	7	162	19	37
14	8	168	21	40
15	9	167	18	40
16	10	166	18	40
15001	14995	13	155	83
15002	14996	14	154	83
15003	14997	16	149	83
15004	14998	22	148	85
15005	14999	29	144	87
15006	15000	33	141	89

Fig 10.7.6: Test data

The training data is written in this format to help “what if analysis”. Now as mentioned in normal K-means implementation we obtain the distance of the each data point (pixel value) from all the assumed centroid point. We now find the minimum, among the distances to all the four clusters, for each data point. The sum of these minimum distances found out for all the data point is found out. [This is actually the sum of distances to the nearest centroid point]. This is our objective function which is to be minimized. The below fig 10.7.7 illustrates the whole process of obtaining the Total distance to be minimized.

	U	V	W	X	Y	Z	AA
6	INDEX	Distance to CENTROID 1	Distance to CENTROID 2	Distance to CENTROID 3	Distance to CENTROID 4	CLASS	Minimum Distance
7	1	69.65503173	246.8885409	179.4010096	135.4297732	Cluster1	69.65503173
8	2	61.55203938	288.2733383	180.7619612	155.9180148	Cluster1	61.55203938
9	3	48.23476514	291.502662	193.7221676	167.7852699	Cluster1	48.23476514
10	4	38.43265276	291.8112275	201.2615501	175.0002603	Cluster1	38.43265276
11	5	38.12198395	296.7798261	202.5874036	178.9067362	Cluster1	38.12198395
12	6	33.03022291	292.6917734	206.0946501	179.4988737	Cluster1	33.03022291
13	7	33.77735709	295.7238411	207.3023232	181.2318632	Cluster1	33.77735709
14	8	29.06818929	291.0147423	210.2986799	181.4713731	Cluster1	29.06818929
15	9	30.16426942	293.3944482	211.3059804	182.5275776	Cluster1	30.16426942
16	10	31.07774291	293.6040996	210.5248167	181.9799732	Cluster1	31.07774291
14996	14990	232.0022672	272.5644391	17.91025256	98.08150969	Cluster3	17.91025256
14997	14991	232.039777	270.949618	19.63830588	97.18821332	Cluster3	19.63830588
14998	14992	232.039777	270.949618	19.63830588	97.18821332	Cluster3	19.63830588
14999	14993	232.039777	270.949618	19.63830588	97.18821332	Cluster3	19.63830588
15000	14994	232.0022672	272.5644391	17.91025256	98.08150969	Cluster3	17.91025256
15001	14995	232.0022672	272.5644391	17.91025256	98.08150969	Cluster3	17.91025256
15002	14996	230.6442481	272.0579487	18.06552506	97.15422221	Cluster3	18.06552506
15003	14997	226.2237482	271.9549456	17.88118468	94.2811055	Cluster3	17.88118468
15004	14998	221.4524899	266.5024079	23.78624865	89.25551164	Cluster3	23.78624865
15005	14999	214.2668055	261.3123923	30.78237886	82.94477111	Cluster3	30.78237886
15006	15000	210.0368808	258.1913542	35.50634084	78.5554043	Cluster3	35.50634084
15007						Sum Of Total Distance	409278.9031

Fig 10.7.7: Total distance to be minimized

Now we utilize solver to find out an optimal value for X and Y for each of the centroids by minimizing sum of, distance to closest centroid points. The optimized values of the centroid points found by minimizing the sum of distances to the nearest centroid points is shown in the below figure 10.7.8.

	H	I	J	K	L	M	N	O	P	Q	R	S
6	Cluster Center 1			Cluster Center 2			Cluster Center 3			Cluster Center 4		
7	R	G	B	R	G	B	R	G	B	R	G	B
8	194.471477	21.33009441	27.9953529	228.0325131	231.109502	232.196969	1.74055439	145.533579	72.7829	66.697	117.177	155.843

Fig 10.7.8: Optimized centroid values

Now we again plot the "Indian Flag" by assigning different colors to the points belonging to different clusters. This is also done with the help of a macro. The clustered image is obtained. **(See attached CD for implementation and color image of the below figure)**.



Fig 10.7.9: Segmented Image

10.7.2 Clustering large cities for sales Analysis

It is not just image pixels that can be clustered. Clustering can be done on any N dimensional dataset because; it's just grouping of data points into K clusters. Here we are considering an application of clustering where we aim at grouping together cities of having same attributes. This comes helpful when it is required to predict the expected sales of a product in these cities. The attributes are selected in such a way that, it is somehow related to the sales of the considered product. For example the below table shows, the demographic data of the largest cities in the United States. For example Atlanda has 67% African American, 2% Hispanic and 1% Asian. It has a median age of 31, 5% unemployment rate, and per capita income of $22000. [In the table column labels we have used some abbreviations to represent these features]. Now one has to group these 49 cities into four clusters that are demographically similar.

TEST DATA						
SL NO	PctAfrAmer	PctHispanic	PctAsian	MedianAge	UnempRate	PCIncome
1	3	35	2	32	5	18
2	67	2	1	31	5	22
3	12	23	3	29	3	19
4	59	1	1	33	11	22
5	26	11	5	30	5	24
6	32	1	2	32	3	20
7	39	20	4	31	9	24
8	38	1	1	31	8	21
9	47	5	1	32	13	22
10	23	1	2	29	3	13
⋮	⋮	⋮	⋮	⋮	⋮	⋮
40	48	1	1	33	8	23
41	7	56	1	30	5	17
42	9	21	12	31	8	20
43	11	14	29	36	6	31
44	5	27	20	30	8	26
45	10	4	12	35	5	28
46	20	4	1	32	6	19
47	4	29	2	31	3	19
48	14	3	1	33	4	20
49	14	3	4	29	6	18

Fig 10.7.10: Demographic data of the largest cities in the United States

As mentioned, our aim is to cluster this demographic data into four clusters. Here we use the K-Means algorithm for clustering. Hence as the first step, we need randomly select the four cluster centers (centroids). Here we have selected four cluster centers since k=4. Lets this be (3, 35, 2, 32, 5, 18), (67, 2, 1, 31, 5, 22), (13, 23, 2, 34, 7, 23) and (12, 23, 3, 29, 3, 19). This is entered in the separate cells as shown in the figure 10.7.11.

Cluster Center 1					
PctAfrAmer	PctHispanic	PctAsian	MedianAge	UnempRate	PCIncome
3	35	2	32	5	18
Cluster Center 2					
PctAfrAmer	PctHispanic	PctAsian	MedianAge	UnempRate	PCIncome
67	2	1	31	5	22
Cluster Center 3					
PctAfrAmer	PctHispanic	PctAsian	MedianAge	UnempRate	PCIncome
13	23	2	34	7	23
Cluster Center 4					
PctAfrAmer	PctHispanic	PctAsian	MedianAge	UnempRate	PCIncome
12	23	3	29	3	19

Fig 10.7.11: Initial Cluster centers

As the next step we need to find out the distance of data points to these assumed cluster centers. To make the calculation easy we use what if analysis to obtain the distances. The figure 10.7.12 shows the prerequisite table for what-if analysis where the provision of index number in the left most column will result in the automatic population of cell values from the corresponding row index of the table shown in fig 10.4.1. This is achieved using the INDEX function. [The functionally of INDEX function is said in detail in the introduction chapters of this book]. E.g. The formula used in second column of the Fig 10.4.3 is =INDEX(C7:C55,$J$19,1) where C7:C55 represents the range over which the table shown in Fig 10.4.1 spans, $J$19 represents the first column of Figure 10.4.3, which is the column index and 1 is the row index.

INDEX OF CURRENT POINT	PctAfrAmer	PctHispanic	PctAsian	MedianAge	UnempRate	PCIncome
1	3	35	2	32	5	18

Fig 10.7.12: Selected training data(here it is1st)

Now we calculate the sum of, mimimum distances of a data point to the four cluster centers. This is calculated for all the data points. Now we find out the sum of these minimum distances calculated and store it in the cell corresponding to the label “TOTAL”. The whole process can be seen in the figure 10.7.13. The different clusters can be easily identified by the color assigned to them.(This coloring is done with the help of conditional formatting in Microsoft excel)

INDEX	Distance to CENTROID 1	Distance to CENTROID 2	Distance to CENTROID 3	Distance to CENTROID 4	CLASS	Minimum Distance
1	41.16454906	71.49361449	15.33684957	31.12654078	Cluster3	159.1215539
2	37.2929723	1.00030557	58.9887278	45.275269	Cluster2	142.5572747
3	26.68862195	58.37904919	0.607228024	16.6930771	Cluster3	102.3679763
4	29.75561587	9.469300946	52.70924455	38.04044002	Cluster2	129.9746014
5	9.904339702	41.55366737	19.30631996	4.912823878	Cluster4	75.67715091
6	5.019928955	34.39181697	30.01463704	14.56255814	Cluster1	83.9889411
7	19.26493441	33.1702746	28.12709933	18.64799971	Cluster4	99.21030805
8	9.124627905	28.37595764	34.62830204	18.77113588	Cluster1	90.90002347
9	18.61680735	20.97332995	40.79761772	26.03021773	Cluster1	106.4179728
10	12.03248616	44.18824328	25.56296496	14.14099004	Cluster1	95.92468444
11	17.59084268	41.1699078	19.15599951	12.56763335	Cluster4	90.48438335
12	25.54389966	57.41323364	7.424942897	15.72964995	Cluster3	106.1117262
13	46.31166597	10.50723807	67.31298456	53.93842742	Cluster2	178.070316
14	71.08505311	92.81161083	47.70434029	61.7427309	Cluster3	273.3437351
15	18.37777444	47.99382011	11.98923922	9.511800329	Cluster4	87.8726341
16	36.76800276	65.67186517	15.92291915	26.25782627	Cluster3	144.6206134
17	75.81093613	96.07658495	71.56109541	70.53525501	Cluster4	313.9838715
18	24.49082978	46.46300937	17.20526681	17.43873626	Cluster3	105.5978422
19	8.493432806	44.25813235	24.77763669	11.19004939	Cluster1	88.71925123
20	5.571088607	41.38739203	24.50070021	9.314878208	Cluster1	80.77405905
21	0.195901422	36.36408812	26.85985777	10.9748043	Cluster1	74.39465162
22	21.13724207	56.49163054	11.23565712	12.22981318	Cluster3	101.0943429
23	28.83258977	58.30735071	11.97888796	18.19328047	Cluster3	117.3121089
24	40.706136	65.56675536	19.82611547	30.72964548	Cluster3	156.8286523
25	25.55230694	11.98938456	48.76930235	34.04371353	Cluster2	120.3547074
26	59.61151945	73.26254749	43.92038309	52.44762734	Cluster3	229.2420774
27	4.115716631	35.51113357	25.83753832	10.39275205	Cluster1	75.85714057
28	17.37177596	53.35919654	21.9288311	13.93753838	Cluster4	106.597342
29	8.580240381	43.42567222	25.66468625	12.04122774	Cluster1	89.71182659
30	32.34417245	6.323371777	53.66426624	39.99487374	Cluster2	132.3266842
31	22.33857997	44.56482243	21.03169764	16.16876716	Cluster4	104.1038672
32	22.80003539	29.57009751	36.31029703	24.14287026	Cluster1	112.8233002
33	14.49711808	50.56014503	19.26542362	10.60429276	Cluster4	94.92697948
34	16.94968308	53.27777512	20.73884352	13.63101324	Cluster4	104.597315
35	11.14107413	27.01611998	33.70931678	18.32881357	Cluster1	90.19532446
36	29.7701878	63.93594135	8.253748185	20.28088674	Cluster3	122.2407641
37	5.446070896	40.50041071	27.23324468	11.67936658	Cluster1	84.85909286
38	22.34654733	58.57617837	21.92563661	17.49313501	Cluster4	120.3414973
39	23.75462952	55.09672685	15.08669402	13.98222783	Cluster4	107.9202782
40	18.62257482	18.62257534	42.877633	27.55426688	Cluster1	107.67705
41	57.04246377	80.37849902	33.3165474	47.66039156	Cluster3	218.3979018
42	29.1886024	61.44698796	10.78724169	18.53802497	Cluster3	119.960857
43	36.66269318	64.08816571	30.66238745	29.39328423	Cluster4	160.8065306
44	39.27599725	69.11353214	20.30827839	28.82285907	Cluster3	157.5206668
45	23.81176187	57.88001869	23.82631363	18.4251071	Cluster4	123.9432013
46	10.08068739	46.36669308	21.32405662	9.007459533	Cluster4	86.77889662
47	36.21334017	67.9884499	10.22932896	26.24912474	Cluster3	140.6802438
48	16.02290519	52.32453853	20.90209872	13.13842619	Cluster4	102.3879686
49	16.87350519	52.49723387	20.60223667	12.95858596	Cluster4	102.9315617
					TOTAL	6022.53175

Fig 10.7.13: What if Datatable/ Total distance calculation

Our objective then is to reduce this distance (which is distance of datapoints to its nearest centroid). As usual we utilize solver to find out an optimal value for X and Y for each of the centroids by minimizing sum of, distance of datapoints to its nearest centroid. The figure 10.7.14 shows the optimized values of cluster centers.

Cluster Center 1					
PctAfrAmer	PctHispanic	PctAsian	MedianAge	UnempRate	PCIncome
29.83686954	3.973552675	1.084752971	32.9383304	5.996066993	20.9919573
Cluster Center 2					
PctAfrAmer	PctHispanic	PctAsian	MedianAge	UnempRate	PCIncome
66.23254211	1.861821803	0.839899268	30.93566864	5.336216139	21.5002857
Cluster Center 3					
PctAfrAmer	PctHispanic	PctAsian	MedianAge	UnempRate	PCIncome
12.17601997	23.29732374	3.375464275	29.18222955	3.257299585	19.0946432
Cluster Center 4					
PctAfrAmer	PctHispanic	PctAsian	MedianAge	UnempRate	PCIncome
22.909491	11.45441243	4.559330982	31.7149606	6.531312035	21.0170641

Fig 10.7.14: Optimal value of cluster centers

With these cluster centers we will be able to group these 49 cities into four clusters that are demographically similar.

10.8 Summary

Data Clustering or grouping together of similar data comes useful in all real world data processing applications. This chapter deals with the implementation of k-Means clustering in Microsoft Excel 2007 with the help of “What-If Analysis” and “Solver” add-in packages available by default. A few applications of clustering like image segmentation and clustering of cities for sales analysis which implemented in Microsoft Excel is also described here. The chapter also explains about a newly developed add-in which can read any image and obtain the pixel values for red, green and blue component of the image.

Questions:

1) What do you understand by data clustering? Where is this utilized and why?
2) Explain the types of clustering and its application domains?
3) What do one refer to as cluster dissimilarity/similarity ?
4) Explain some application of data clustering?
5) What is Image Segmentation? Implement segmentation of an image in Microsoft Excel?

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